

Report and Case Study on Modern Graphics Systems

You are required to complete two parts of this assignment.

Part A - Report

Write a report on the latest components of a graphics system and modern input devices used in graphics.

Part B - Case Study

Select a real-world modern graphics system, device, or environment and present it as a case study. Your case study should explain how the graphics system components and input devices are used in practice.

Part A - Report

Write a structured report explaining the following,

1. Modern Components of a Graphics System

Discuss the role and importance of modern graphics system components such as,

- CPU
- GPU
- RAM and VRAM
- Frame Buffer
- Display Controller
- Graphics Card
- Modern Display Units

2. Modern Input Devices Used in Graphics

Discuss modern graphics input devices such as,

- Graphics tablets
- Stylus / digital pen
- Pen displays
- Touchscreens
- 3D scanners
- Cameras used for graphics capture
- VR/AR controllers
- Eye-tracking devices
- Gesture-based devices

3. Importance in Modern Applications

Explain how these components and devices are used in areas such as,

- Digital drawing
- Animation
- Gaming
- 3D modelling
- Virtual reality
- Augmented reality
- Simulation
- Computer-aided design

Part B - Case Study

Choose one modern graphics-related system or setup and analyze it as a case study. You may choose one of the following, or a similar example,

- A digital artist's workstation
- A gaming PC with a dedicated GPU
- A VR design or training system
- A 3D animation studio setup
- An architect's CAD workstation
- A tablet-based graphics system such as iPad + stylus
- A 3D scanning and modelling environment

Your case study should include,

- Introduction to the selected system
- Description of the graphics system components used
- Description of the input devices used
- Why these components and devices are suitable
- Advantages of the selected setup in real-world use
- Challenges or limitations, if any

Instructions

- Complete both the report and the case study.
- Write in your own words.
- Use clear headings and subheadings.
- Include diagrams, tables, or images where appropriate.
- Use relevant real-world examples.
- Add references for any sources used.